MILITARY SPECIFICATION

GYRO, DIRECTIONAL CN-998( )/ASN-43

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of directional gyroscope which provides the capability of slaving the gyro to the desired magnetic reference, and is designated CN-998( )/ASN-43 (See 6.1 and 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	Box, Fiberboard
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper (For Carton Sealing)

MILITARY

MIL-P-116	Preservation, Methods of
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-P-26514	Polyurethane Foam, Rigid, or Plastic, for Packaging

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-252	Wired Equipment Classification of Visual and Mechanical Defects for
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Requirements for Equipment

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MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods

DRAWINGS

USAECOM

SC-DL-545357	Gyro, Directional CN-998( )/ASN-43
SM-B-498802	Test Specification

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Description - The CN-998( )/ASN-43 is a case level gyro in which the gyro assembly (inner gimbal) maintains a level, or upright position within the gimbal ring (outer gimbal). Changes in aircraft heading are reflected by corresponding changes in the position of the inner gimbal with respect to the outer gimbal. These changes in relationship between the inner and outer gimbals produce stablized reference output signals which are routed to an indicator or compass controller for a visual indication of aircraft heading.

3.2 First Article - When specified in the contract, the contractor shall furnish first article units in accordance with 4.3.

3.3 Construction - The equipment shall be constructed in accordance with the requirements of this specification and of the following drawing (including the requirements of parts, materials and processes thereon): SC-DL-545357.

3.4 Cleaning -

3.4.1 Parts - After fabrication, parts shall be cleaned in accordance with Requirement 9 of MIL-STD-454, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

3.4.2 Units - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

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3.5 Finish - The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings (See 4.4).

3.6 Securing of Parts -

3.6.1 General - Brackets, lugs, inserts, bolts and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.6.2 Securing of Parts by Threaded Fasteners - Nylok or other screws and nuts with plastic devices and Loctite or similar sealants shall not be subjected to temperatures in excess of 250°F(as during baking of paint). Loctite or similar sealants shall be applied in accordance with the manufacturer's instructions.

3.7 Marking -

3.7.1 General - Marking shall conform to MIL-M-13231 and the equipment drawings (See 4.4)

3.7.2 Visibility - Whenever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.7.3 Serial Numbers - The unit(s) on this order shall have serial numbers.

3.8 Soldering -

3.8.1 Solder - Solder shall conform to Requirement 5 of MIL-STD-454 and the equipment drawings.

3.8.2 Acid or Acid Salts - No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.8.3 Process - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystalized, overheated or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in

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the equipment; flux being spread over a larger area; or damage to the unit. Insulation material that has been subjected to heating during the soldering operation shall be undamages and parts fastened thereto shall not have become loosened.

3.8.4 Clearance - Clearance between solder connections or bare conductors, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.9 Connections -

3.9.1 General - Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.10 Tropicalization of Materials - Materials shall be treated as follows:

3.10.1 Treating Materials - Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.10.2 Toxicity - Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabriation, transportation, operation or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.10.3 Flexibility - Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.10.4 Statement of Treatment - When required to be submitted by the contract, the statement of treatment shall describe in detail the materials to be treated and the treating materials and processes that the contractor proposes to use.

3.11 Reliability - The equipment shall have a mean-time-between failure (MTBF) of 1000 hours when subjected to the following condition (See 4.5.6).

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Temperature: -54°C to +55°C (-65°F to 131°F)
Duty Cycle: 100%
Environment: Airborne

3.12 Service Conditions(See 4.7) - The units(s) shall meet the following service conditions:

3.12.1 Temperature-Altitude -

a. Operating - The equipment shall be capable of continuous operation at altitudes up to 70,000 feet above sea level and at temperatures ranging from -54°C to +71°C. Intermittent operation to +95°C (See 4.7.1).

b. Non-Operating - The equipment shall be capable of withstanding exposure at altitudes up to 70,000 feet above sea level at temperatures ranging from -62°C to +95°C (See 4.7.1).

3.12.2 Vibration - The equipment shall be capable of withstanding vibration encountered in both airplanes and helicopters(See 4.7.2).

3.12.3 Humidity - The equipment shall be capable of withstanding exposure to relative humidities of 95% or greater(See 4.7.3).

3.12.4 Salt Fog - The equipment, when subjected to salt fog, shall show no evidence of degradation, such as flaking, pitting, blistering or, loosening of finish or metal surface; or exfoliation of metal(See 4.7.4 and 6.5).

3.12.5 Explosive Conditions - The equipment shall not cause ignition of an ambient-explosive-gaseous mixture with air when operating in such an atmosphere(See 4.7.5).

3.12.6 Dust - The equipment shall withstand, in both an operating and non-operating condition, exposure to dust particles as encountered in operational areas of the world(See 4.7.6).

3.12.7 Fungus - The equipment shall withstand exposure to fungus growth as encountered in tropical climates(See 4.7.7).

3.12.8 Shock - The equipment (with vibration isolators in place, if any) shall not suffer damage or subsequently fail to provide the performance specified herein when subjected to shock(See 4.7.8).

3.12.9 Crash Safety - The equipment and mounting fixture shall be capable of withstanding shock(crash safety) as specified herein(See 4.7.9 and 6.4)

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3.13 Electrical Requirements -

3.13.1 Starting Time - The directional gyroscope shall be ready for operation three minutes after power is applied (gyroscope wheel speed of 18,000 rpm minimum) (See 4.8.1).

3.13.2 Primary Input Power - The directional gyroscope shall meet all applicable requirements and shall give specified performance when operated from an AC power source with characteristics and limits as defined in MIL-STD-704. The AC power requirement shall be 115 volts, 440 Hz, single phase, Category "B", 0.260 amperes maximum(See 4.8.2).

3.13.2.1 Power Consumption - The electrical power consumption of the directional gyroscope shall not exceed 60 VA starting and 50 VA running(See 4.8.3).

3.13.3 Synchronizing Signal Outputs - The annunciator output(pins "a" and "b" of Connector J1) shall be 150 plus or minus 50 microamperes for a five degree changes in magnetic heading from a null (synchronized) position. The current indication shall be positive(See 4.6.8) for a clockwise heading changes of the induction compass transmitter, and negative for a counter-clockwise change. The heading transmitter output(pins "G", "H" and "J" of Connector J1) shall show an angular heading change in the same direction as the magnetic heading change of the induction compass transmitter when the gyroscope is synchronized to the new heading(See 4.8.4).

3.13.4 Slaving Rate - The gyroscope shall precess at a rate of 2.5 plus or minus 1.0 degree per minute when the magnetic heading is changed by thirty degrees(See 4.8.5).

3.13.5 Static Settling - The output of the heading transmitter, T-611 shall be within plus or minus 1.0 degree of the magnetic heading input for a ten degree clockwise and counter-clockwise change in the directional gyroscope heading position, after the synchronized condition is attained. The output angular readings shall not differ by more than 0.25 degrees from each other(See 4.8.6).

3.13.6 Dynamic Accuracy - When the induction compass transmitter, T-611 is subjected to the dynamic conditions of 4.8.7, the heading transmitter output shall be within plus or minus 1.5 degree of the magnetic heading input for a ten degree clockwise and counter-clockwise change in the directional gyroscope heading position, and the individual output readings shall not differ by more than 0.5 degrees(See 4.8.7).

3.13.7 Scale Error - The scale error(difference between mangetic heading input and synchronized heading transmitter output) of the directional gyroscope shall not exceed 0.75 degrees for any heading(See 4.8.8).

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3.13.8 Drift - The gyroscope free drift rate shall not exceed 5.5 degrees per hour for any heading, when corrected to test site latitude. For test site latitude condition, where four magnetic headings are checked, the numerical average of the four heading drift rates shall not exceed 5.5 degrees per hour and only one individual heading drift rate may exceed 5.5 degrees per hour but shall be not more than 6.5 degrees per hour(See 4.8.9).

3.13.9 Radio Frequency Interference(See 4.8.10) - The Directional Gyroscope CN-998( )/ASN-43 shall meet the emanation and susceptibility test of 4.8.10.

3.14 Burn-In/Debugging - Each unit on order shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the burn-in/debugging as specified (See 4.5.1).

3.15 Interchangeability - Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment(See 4.9). Individual items shall not be hand-picked for fit or performance unless selective fit is indicated on individual drawings; matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.16 Weight - The weight of the CN-998( )/ASN-43 Directional Gyro shall not exceed 5.8 pounds(See 4.11).

3.17 Workmanship - The units(s) shall be manufactured and assembled in accordance with the equipment drawings, Requirement 9 of MIL-STD-454, and the applicable portions of the following paragraphs herein:

Cleaning	3.4
Securing of Parts	3.6
Soldering	3.8
Connections	3.9
Tropicalization of Materials	3.10

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of Inspection - Inspection shall be classified as follows:

a. First Article Inspection(does not include preparation for delivery).

b. Inspection covered by subsidiary documents(See 4.4).

c. Quality Conformance Inspection.

(1) Quality conformance inspection of equipment before preparation for delivery(See 4.5).

(2) Quality conformance inspection of preparation for delivery(See 4.12 and 4.13).

4.3 First Article - Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor.

4.3.1 First Article Inspection - The first article inspection shall consist of inspections specified in Table I, and shall be performed in the order specified therein:

TABLE I - FIRST ARTICLE INSPECTION

INSPECTIONS	REQT PARA	TEST PARA	SAMPLE UNITS
Inspections covered by subsidiary documents	-	4.4	All Units
Burn-In/Debugging	3.14	4.5.1	All Units
Group A	(See Table II)		All Units
Group B	(See Table III)		All Units
Group C	(See Table IV)		One half for Sub Group 1 One Half for Sub Group 2
Reliability	3.11	4.5.6	3 Units

(NOTE: The inspections 1 through 4, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table. After completion of Group C inspections, conforming units shall be reinspected and shall pass all Group A inspection.)

4.3.2 First Article Data - The first article test plan and test report(s) shall be as required in the contract.

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4.4 Inspection Covered by Subsidiary Documents - The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order:

ITEM	WHERE REQUIRED
Finish	3.5
Marking	3.7

4.5 Quality Conformance Inspection of Equipment Before Preparation for Delivery - The contractor shall perform the inspection specified in 4.4 and 4.5.1 thru 4.5.6. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specified requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified inspection to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Burn-In/Debugging - All units presented to the Government for acceptance shall be Burned-In in accordance with Drawing SM-B-498802.

4.5.2 Group A Inspection - Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table II. Each lot shall be subjected to verification utilizing the procedures of MIL-STD-105, using the general inspection levels and the AQL's indicated in Table II. Group A inspection shall be performed in any order Which is satisfactory to the Government.

TABLE II - GROUP A INSPECTION

INSPECTION	REQT PARA	INSP PARA	AQL MAJOR	AQL MINOR
Visual and Mechanical	3.17	4.10	1%	4%
Synchronizing Signal Outputs	3.13.3	4.8.4	1.0%	*
Slaving Rate	3.13.4	4.8.5	For the Group	
Static Settling	3.13.5	4.8.6		
Scale Error	3.13.7	4.8.8		
Drift	3.13.8	4.8.9	2.5%	*

* All electrical defects are considered major.

4.5.3 Group B Inspection - Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group inspection. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105. The inspection level shall be S-3.

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TABLE III - GROUP B INSPECTION

INSPECTION	REQT PARA	INSP PARA	AQL
Interchangeability	3.15	4.9	6.5%
Starting Time	3.13.1	4.8.1	
Primary Input Power	3.13.2	4.8.2	6.5%
Power Consumption	3.13.2.1	4.8.3	For the
Dynamic Accuracy	3.13.6	4.8.7	Group

4.5.3.1 Order of Inspection Within Group B - Group B inspection shall be performed in any order which is satisfactory to the Government. Each sample shall be subjected to all tests; separate sampling shall not be permitted.

4.5.4 Group C Inspection - This inspection shall consist of the tests specified in Table III and shall be performed on sample units that have been selected in accordance with 4.5.4.1.

TABLE IV - GROUP C INSPECTION

INSPECTION	REQT PARA	INSP PARA
Sub-Group 1(Note 1)		
Temperature-Altitude	3.12.1	4.7.1
Vibration	3.12.2	4.7.2
Dust	3.12.6	4.7.6
Humidity	3.12.3	4.7.3
Salt Fog	3.12.4	4.7.4
Shock	3.12.8	4.7.8
Sub-Group 2(Note 1)		
Explosive Conditions	3.12.5	4.7.5
Radio Frequency Interference	3.13.9	4.8.10
Crash Safety	3.12.9	4.7.9
Fungus	3.12.7	4.7.7

(Note 1: To be performed in the order shown.)

4.5.4.1 Sampling for Group C Inspection - Sample units shall be selected at random for each subgroup of Table IV.

4.5.4.1.1 Sub-Group 1 - Two units shall be selected from the first production lot. For subsequent Group C inspection, two units shall be selected from every two months production or from 80 units, whichever occurs first.

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4.5.4.1.2 Sub-Group 2 - One unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected from every four months production or from 320 units, whichever occurs first.

4.5.5 Reinspection of Group C Sample Units - Unless otherwise specified, sample units which have been subjected to and passed inspection may be accepted on contract or order provided the sample units are refurbished and are resubjected to and pass Group A inspection.

4.5.6 Reliability Acceptance Testing - The contractor shall conduct a reliability acceptance test on every lot produced in accordance with Test Plan V, Test Level E of MIL-STD-781 using a specified MTBF of 1000 hours. A lot is defined as one month's production. The sample size shall be a minimum of eight(8) equipments(See 3.11).

4.5.6.1 Failure Definition - A failure is defined as any inability of the equipment to perform its required function, including any departure from performance specified in paragraph 3.13.

4.6 Test Conditions -

4.6.1 Normal Conditions - Unless otherwise specified, all inspection required by this technical requirement shall be made under the following conditions:

a. Temperature - Ambient room conditions.

b. Pressure - Ambient room conditions.

c. Humidity - Ambient room conditions.

d. Input Power - 115 plus or minus 2 volts, ac, 400 plus or minus 10 Hz, single phase.

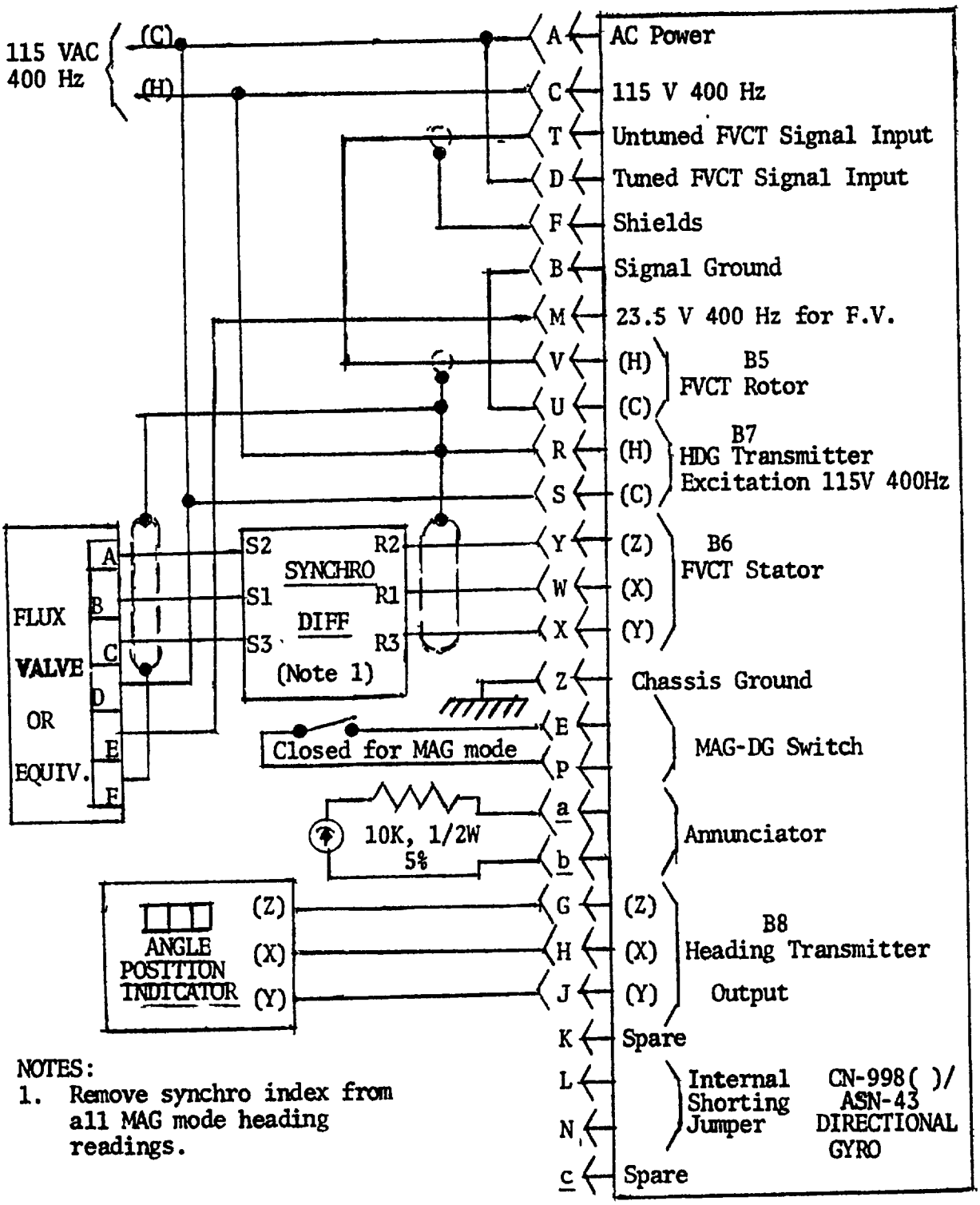
4.6.2 Attitude - Unless otherwise specified, the CN-998( )/ASN-43 shall be tested in its normal operating position.

4.6.3 Interconnection - Unless otherwise specified, all tests required by this technical requirement shall be conducted with the CN-998( )/ASN-43 interconnected as shown in Figure 1.

4.6.4 Test Reading - Unless otherwise specified, all azimuthal outputs from the directional gyroscope shall be displayed on an angle position indicator having a minimum input impedance of 400 ohms.

4.6.5 Magnetic Reference - Unless otherwise specified, all slaved mode (MAG) tests required by this specification shall be conducted using a simulated induction compass transmitter source in an area where the earth's magnetic field is approximately 0.18 oersted horizontal, and 0.54 oersted vertical. This field shall be maintained at a constant magnitude and direction during tests(See Figure 1).

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NOTES:
1. Remove synchro index from all MAG mode heading readings.

FIGURE 1. Test Interconnection Diagram for CN-998( )/ASN-43

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4.6.6 Scorsby Table - All drift, slaving rate, and settling error tests specified herein shall be performed with the CN-998( )/ASN-43 Directional Gyro mounted on a reversing 2.5 degree total excursion Scorsby(Table, Tilting, Gyro Instrument Testing MX-4042A/ASW-12, or functional equivlent) which has a frequency of 5 to 7 cycles per minute. The Scorsby shall be initially leveled within plus or minus 0.25 degrees. All readings shall be taken with the Scorsby rotating in the same direction and at the same extreme of excursion, either high or low, unless othewise specified.

4.6.7 Operating Mode - Unless otherwise specified, the directional gyro shall be operated in the slaved(MAG) mode.

4.6.8 Null Meter - The null meter(annuciator) shown in Figure 1 shall be a 500-0-500 microampere DC current meter having an accuracy of two percent of full scale or better. The polarity of the meter shall be established by applying a positive DC voltage to one terminal of the meter with respect to the other. The terminal with the positive voltage which causes a needle deflection to the left shall be connected to PIN "A" of Connector J1.

4.6.9 Warmup Time - A five minute warmup will be allowed before conducting any test unless otherwise specified.

4.7 Service Conditions - The unit(s) shall be subjected to and meet the service conditions in accordance with MIL-STD-810 except as defined herein.

4.7.1 Temperature-Altitude Tests(See 3.12.1) - The directional gyroscope shall be subjected to the temperature and altitude test conditions of Method 504 for Class 2 equipment of MIL-STD-810. Operational and performance checks will be made as specified in the temperature and altitude test procedure by performing and determining compliance with the following tests; 4.8.6, except that the two heading readings shall not differ by more than 0.4 degrees; 4.8.8 and 4.8.9.1, for one heading only and except that drift shall not exceed 4.75 degrees(9.5 degrees per hour).

4.7.2 Vibration Test(See 3.12.2) - The directional gyrscope shall be tested in accordance with the vibration tests of MIL-STD-810, Method 514-1, Procedure I, Curve Z, Part I. During each mode of resonance vibration, the gyroscope shall be subjected to and meet 4.8.8, except the number of test headings shall be limited to one. During and subsequent to the vibration cycling test, the directional gyroscope shall be subjected to and meet 4.8.8 and 4.8.9.1, except the number of test headings shall be limited to one during each test, and the tolerance for 4.8.9.1 shall be 4.75 degrees(9.5 degrees per hour).

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4.7.3 Humidity Test(See 3.12.3) - The directional gyroscope shall be tested in accordance with the humidity test of Method 507, Procedure II, MIL-STD-810, including measurements at the 24th hour. After completion of test, the gyroscope shall meet 4.8.6, 4.8.8 and 4.8.9.1 of this specification, except the number of test headings for 4.8.9.1 shall be limited to one and the drift tolerance shall be 2.75 degrees.

4.7.4 Salt Fog Test (See 3.12.4) - The equipment(including accessories) in its normal operating position, shall be tested in accordance with Method 509, Procedure 1 of MIL-STD-810. The equipment shall not be opened. Operation is not required. No rust shall be imposed on the test item by the test facility. After the completion of the test, the gyroscope shall meet 4.8.6, 4.8.8 and 4.8.9.1 of this specification, except the number of test headings for 4.8.9.1 shall be limited to one and the drift tolerance shall be 2.75 degrees.

4.7.5 Explosion Condition(See 3.12.5) - The directional gyroscope shall be tested in accordance with the explosion test of MIL-STD-810, Method 511, Procedure 1.

4.7.6 Dust Test(See 3.12.6) - The equipment shall be tested in accordance with Method 510, Procedure 1 of MIL-STD-810. After the completion of the test, the gyroscope shall meet 4.8.6, 4.8.8 and 4.8.9.1 of this specification, except the number of test headings for 4.8.9.1 shall be limited to one and the drift tolerance shall be 2.75 degrees.

4.7.7 Fungus Test(See 3.12.7) - The equipment(including accessories) shall be tested in accordance with Method 508, Procedure 1 of MIL-STD-810. Spraying shall include parts and materials on and under the chassis, and the interior of the equipment case. The equipment shall remain in an open condition during the test. After the fungus test, the equipment shall be visually examined, using a 10 power magnifier. The equipment shall show no more than six minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growth as evidenced by growth colonization(which includes branching and sporulation on or within each cubic foot, or fraction thereof, of equipment assembly volume). Isolated instances of partial tubular germination shall no be included in this evaluation. Operation is not required during the test. After the completion of the test, the gyroscope shall meet 4.8.6, 4.8.8 and 4.8.9.1 of this specification, except the number of test headings for 4.8.9.1 shall be limited to one and the drift tolerance shall be 2.75 degrees.

4.7.8 Shock Test(See 3.12.8) - The directional gyroscope shall be tested in accordance with the shock test equipment procedure of MIL-STD-810, Method 516.1, Procedure 1. After the completion of the test, the gyroscope shall meet 4.8.6, 4.8.8 and 4.8.9.1 of this specification, except the number of test headings for 4.8.9.1 shall be limited to one and the drift tolerance shall be 2.75 degrees.

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4.7.9 Crash Safety Test(See 3.12.9) - The directional gyroscope shall be tested in accordance with the shock test, mounting(crash safety) procedure of MIL-STD-810, Method 516.1, Procedure III.

4.8 Electrical -

4.8.1 Starting Time(See 3.13.1) - With the gyroscope wheel stopped (non-rotating), power shall be applied to the directional gyroscope and the wheel speed(in rpm) measured at the end of three minutes. Rotational speed of the gyro wheel shall be 18,000 rpm minimum.

4.8.2 Primary Input Power Tests(See 3.13.2) - Power shall be applied to the directional gyroscope and tests run at the following combination of voltage and frequency:

a. 107.5 plus or minus 1 volts at 380 plus or minus 2 Hz.

b. 107.5 plus or minus 1 volts at 420 plus or minus 2 Hz.

c. 119.5 plus or minus 1 volts at 380 plus or minus 2 Hz.

d. 119.5 plus or minus 1 volts at 420 plus or minus 2 Hz.

The directional gyroscope shall meet the tests specified in 4.8.6, 4.8.8 and 4.8.9.1(for test site latitude only and at one test heading).

4.8.3 Power Consumption Test(See 3.13.2.1) - The total power required for the CN-998( )/ASN-43 Directional Gyro when measured as shown in Figure 1 shall not exceed 60 VA, starting, and 50 VA running. The running power shall be measured after a minimum of five minutes after application of power.

4.8.4 Synchronizing Signal Output Test(See 3.13.3) - The synchronizing signal outputs shall be determined in the following manner:

4.8.4.1 The case of the directional gyro shall be rotated until the null meter(annunciator) indicates a null when the induction compass transmetter is set to a North(0 degree) heading. The angle position indicator should indicate approximately 0 degrees.

4.8.4.2 Rotation of the transmitter five degrees clockwise from the synchronized position shall cause a deflection of the null meter to the left and an indication of 150 plus or minus 50 microamperes on the null meter. The case of the directional gyroscope shall be rotated clockwise to again null the null meter. The angle position indicator shall indicate a clockwise reading(approximately five degrees).

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4.8.4.3 The induction compass transmitter shall be re-positioned to a North heading and the directional gyroscope synchronized as indicated by the null meter. Rotation of the transmitter five degrees counterclockwise from the synchronized position shall cause a deflection of the null meter to the right and an indication of 150 plus or minus 50 microamperes on the null meter. The case of the directional gyro shall be rotated counterclockwise to again null the null meter. The angle position indicator shall indicated a counterclockwise reading(approximately 355 degrees).

4.8.5 Slaving Rate Test(See 3.13.4) - With the directional gyroscope first synchronized on a North (0 degree) heading, rotation of the induction compass transmitter to a 30 degree heading shall cause the gryoscope to process toward the new transmitter heading. The precession rate shall be 2.5 plus or minus 1.0 degrees per minute when measured over a 10 degree heading change from 355 to 345 degrees.

4.8.6 Static Settling Test(See 3.13.5) - With the induction compass transmitter positioned to a North (0 degree) heading, the case of the directional gyroscope shall be rotated to produce a reading of ten degrees on the angle position indicator. Allow the directional gyro to slave to null condition, and after ten minutes, note the heading indication(low and high excursion reading taken and averaged). Repeat the above, except the gryoscope shall be rotated to produce a reading of 350 degrees on the angle position indicator. The two average heading indications, after settling, shall be 0 plus or minus 1.0 degree(after removal of induction compass transmitter and angle position indicator error) and shall not differ by more than 0.25 degrees.

4.8.7 Dynamic Accuracy Test(See 3.13.6) -The induction compass transmitter shall be subjected to dynamic conditions to produce an oscillatory output signal having a peak amplitude equivalent to plus or minus 7.5 degrees error and a frequency of 2 to 5 Hz. With the induction compass transmitter subjected to the above conditions, the test of 4.8.6 shall be conducted, except the tolerances shall be plus or minus 1.50 and 0.5 degrees.

4.8.8 Scale Error Test(See 3.13.7) - The Scorsby shall be turned off during this test. Record the error between the induction compass transmitter heading and the angle position indicator reading at 30-degree heading increments starting at 0 degrees. The gyroscope shall be resynchronized on each heading by rotating the gyroscope case. The maximum error on any heading(after removal of induction compass transmitter and angle position indicator errors) shall be 0.75 degrees.

4.8.9 Drift Test(See 3.13.8) - All gyroscope drift tests shall be conducted with the MAG-DG switch in the DG position. The gryoscope drift at each of the four headings shall be computed as follows:

$$\text{Drift Angle} - I\text{-}7.5\ (\sin L' - \sin L)$$

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Where I is the change of indication of the angle position indicator; L' is the latitude set on the latitude dial; and L is the actual latitude of the test location.

4.8.9.1 Test Site Latitude Drift - The latitude dial on the base of the CN-998( )/ASN-43 Directional Gyro shall be set to the latitude of the location at which the test is being conducted. The North-South hemisphere switch shall be set to the hemisphere of the test location. After operating the directional gyroscope for a minimum of thirty minutes, the drift shall be determined by recording the heading indication on the angle position indicator, waiting thirty minutes and again recording the heading indication on the angle position indicator. Repeat the above drift test for each of three successive 90 degree spin axis headings beginning 90 degrees upscale from the original heading. (NOTE: Change in spin axis must be accomplished by slaving gyroscope. Rotation of case does not change spin axis heading). The numberical average of the four computed drift readings shall not exceed 2.75 degrees and only one reading may exceed 2.75 degrees bur shall not exceed 3.25 degrees.

4.8.9.2 N 90^0 Latitude Drift - The drift shall be determined as specified in paragraph 4.8.9.1 except the latitude dial shall be set at 90 degrees, the hemisphere switch set to North and the number of test headings shall be limited to one. The computed drift reading shall not exceed 2.75 degrees.

4.8.9.3 S 90^0 Latitude Drift - The drift shall be determined as specified in paragraph 4.8.9.1, except the latitude dial shall be set at 90 degrees, the hemisphere switch set to South, and the number of test headings shall be limited to one. The computed drift reading shall not exceed 2.75 degrees.

4.8.10 Radio Frequency Interference(See 3.13.9) - The CN-998/ASN-43 Directional Gyroscope shall be tested in accordance with the provisions on emanation and susceptibility requirements of MIL-STD-461 and MIL-STD-462.

4.9 Inspection for Dimensional Interchangeability - The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.15. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers and mountings.

d. Location of connectors, locking pins, fasteners, slides and mountings which recieve mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

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e. Size and form of special threads.

4.10 Visual and Mechanical Inspection(See 3.17) - Inspection for visual and mechanical defects shall include, but not be limited to, inspection for the defects listed in the applicable sections of MIL-STD-252, and the equipment drawings, at all points of manufacture.

4.11 Weight(See 3.16) - The CN-998( )/ASN-43 shall be weighed on a suitable balance scale with an accuracy of plus or minus 2 percent of the weight to be measured.

4.12 Quality Conformance Inspection of Preparation for Delivery - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.13 Rough Handling Test(Preparation for Delivery) - When rough handling test in accordance with Specification MIL-P-116 is required by the contract(See 6.2(g)), the following functional tests shall be conducted to determine freedom from operational malfuction caused by the rough handling:

a. Static Settling - 4.8.6

b. Scale Error - 4.8.8

c. Drift at Test Set Latitude - 4.8.9.1

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging - Preservation and packaging shall be Level A or C, as specified(See 6.2(b)).

5.1.1 Level A -

5.1.1.1 Cleaning - Directional Gyro CN-998( )/ASN-43 shall be cleaned in accordance with Process C-1 of MIL-P-116.

5.1.1.2 Drying - Directional Gyro CN-998( )/ASN-43 shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation Application - None required.

5.1.1.4 Unit Packaging - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116. Directional Gyro CN-998( )/ASN-43 shall be Packaged Method III as follows: Cushion each directional gyroscope on all surfaces with polyurethane conforming to MIL-P-26514, designed to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned uchioned gyroscope within a close-fitting box conforming to PPP-B-636, W5c, weather-resistant. Closure shall be made by pressure sensitive tape conforming to PPP-T-76.

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5.1.2 Level C - Directional Gyro CN-998( )/ASM-43 shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing - Packing shall be Level A, B or C as specified(See 6.2(b)).

5.2.1 Level A - A quantity of Directional Gyro CN-998( )/ASN-43 packaged as specified in 5.1, shall be packed in a close fitting fiberboard box conforming to PPP-B-636, Type CF, class weather-resistant W5c; closure shall be in accordance with the box specification.

5.2.2 Level B - A quantity of Directional Gyro CN-998( )/ASN-43, packaged as specified in 5.1, shall be packed in accordance with 5.2.1.

5.2.3 Level C - A quantity of Directional Gyro CN-998( )/ASN-43, packaged as specified in 5.1.2, shall be packed in shipping containers in a manner that will afford adequate protection against damage to the package and its contents during shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation.

5.3 Marking - In addition to any marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use - The CN-998( )/ASN-43 Directional Gyro, covered by this technical requirement, is intended for use as part of the AN/ASN-43( ) Gyromagnetic Compass Set. It provides stabilized reference information to various radiomagnetic course indicators, autopilot navigational equipment, and other auxiliary systems in fixed and rotary wing aircraft.

6.2 Ordering Data - Procurement documents should specify the following:

a. Title, number and date of this document and any amendment thereto.

b. Level of packaging and level of packing required for shipment (Level A, Level B or Level C) and such additional marking that may be required.

c. The specific paragraphs of Section 5 which are applicable to the particular procurement.

d. First Article Inspection(including order of testing, if required).

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e. Marking and shipping of samples.

f. Place of final inspection.

g. Rough handling test(preparation for deliver), if required.

6.3 Nomenclature - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: CN-998W/ASN. The contractor should apply for nomenclature in accordance with the applicable clause in the contract(See 1.1).

6.4 Crash Safety - For the purpose of the crash safety test(See 3.12.9 and 4.7.9), a non-operative gyroscope assembly or an equivalent dummy mass simulating the gyroscope weight distribution and mounting, may be utilized.

6.5 Definitions -

6.5.1 Branching - Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament(hypha).

6.5.2 Growth Colonization - Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growht.

6.5.3 Microbial Growth - Microbial growth is the growth of very minute organisms. Such orginisms when present in large numbers may provide a colony visible to the naked eye.

6.5.4 Sporulation - Sproulation is the formation of minute unicellular reproductive or dormant bodies called spores.

6.5.5 Tubular Germination - Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.5.6 Exfoliation - Exfoliation is corrosion along the grain boundaries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

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SPECIFICATION ANALYSIS SHEET

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Budget Bureau No 22-R255

INSTRUCTIONS. This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION MIL-G-45279(EL) GYRO, DIRECTIONAL CN-998( )/ASN-43

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING.

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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